

Q1. Two signals $s_1(t) = -d \cos(2\pi f_0 t)$, $s_2(t) = 2d \cos(2\pi f_0 t)$, where $0 \leq t \leq T_s$, are used as the signal set for digital transmission in an AWGN channel with noise variance $\frac{N_0}{2}$. T_s is the symbol duration.

$$n \sim \mathcal{CN}(0, \frac{N_0}{2})$$

(a) Indicate the basis function(s) of the signal set and their vector representations under this set of basis function(s). (4 marks)

(b) Find the average bit energy and the signal-to-noise ratio $\gamma_b = \frac{E_b}{N_0}$. (4 marks)

(c) Assume $s_1(t)$ and $s_2(t)$ are sent with equal probability. Derive the decision metric based on the ML rule. (6 marks)

(d) Evaluate the bit error rate and express it using Q function and γ_b . (6 marks)

(a) GSO: $E_2(t) = \int_0^{T_s} s_2^2(t) dt = \frac{T_s}{2} \cdot 4d^2 = 2T_s d^2$

$$\phi(t) = \frac{s_2(t)}{\sqrt{E_2}} = \frac{2d \cos(2\pi f_0 t)}{d \sqrt{2T_s}} = \sqrt{\frac{2}{T_s}} \cos(2\pi f_0 t)$$

$$\vec{s}_2 = \sqrt{E_2} = \sqrt{2T_s} d, \quad \vec{s}_1 = -\sqrt{\frac{T_s}{2}} d$$

(b) $E_1 = \frac{T_s}{2} d^2, \quad E_2 = 2T_s d^2$

(b) bit: $E_b = E_s = \frac{1}{2} (E_1 + E_2) = \frac{5}{4} T_s d^2$

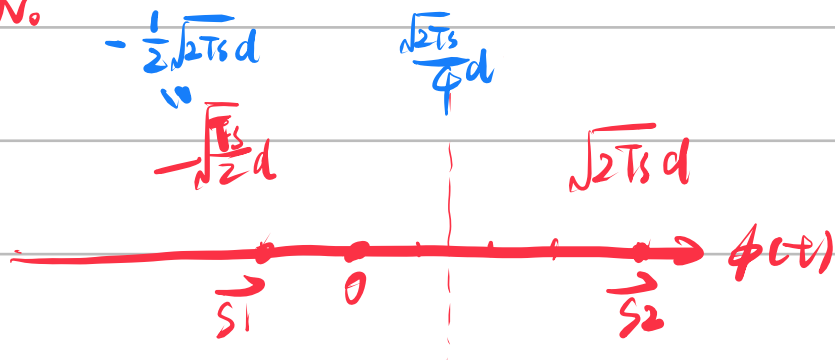
$$\gamma_b = \frac{E_b}{N_0} = \frac{5T_s d^2}{4N_0}$$

$$(\sqrt{\frac{1}{2}} + \sqrt{2}) / 2 = \frac{\sqrt{2}}{4}$$

method 1:

(c) constellation

method 2:



ML rule: $\hat{s} = \arg \max_{1 \leq m \leq M} (P(r|s_m) \underline{P(s_m)})$ MAP

$$\hat{s} = \arg \max (P(r|s_m))$$

$$P(\vec{r} | \vec{s}_1) = P(\vec{n} = \vec{r} - \vec{s}_1) = \frac{1}{\sqrt{2\pi}N_0} e^{-\frac{(r + \sqrt{\frac{T_3}{2}}d)^2}{N_0}}$$

$$P(\vec{r} | \vec{s}_2) = \frac{1}{\sqrt{2\pi}N_0} e^{-\frac{(r - \sqrt{\frac{T_3}{2}}d)^2}{N_0}}$$

$$= e^{\frac{(r - \sqrt{\frac{T_3}{2}}d)^2 - (r + \sqrt{\frac{T_3}{2}}d)^2}{N_0}} \begin{cases} > 0 \rightarrow \vec{s}_1 \\ < 0 \rightarrow \vec{s}_2 \end{cases}$$

$$(r - \sqrt{\frac{T_3}{2}}d)^2 - (r + \sqrt{\frac{T_3}{2}}d)^2 = \frac{3}{2}T_3 d^2 - 2rd(\sqrt{\frac{T_3}{2}} + \sqrt{\frac{T_3}{2}})$$

$$= d\sqrt{T_3} \left[\frac{3}{2}\sqrt{T_3}d - \underbrace{2r(\sqrt{2} + \sqrt{2})}_{-3\sqrt{2}r} \right]$$

$$r \underset{\vec{s}_2}{\overset{\vec{s}_1}{>}} \frac{3}{2}\sqrt{T_3}d \cdot \frac{1}{3\sqrt{2}} = \frac{\sqrt{2}}{4}\sqrt{T_3}d = \frac{\sqrt{2T_3}}{4}d$$

$$-\sqrt{\frac{T_3}{2}}d$$

$$(d) P(e | \vec{s}_1) = P(\vec{r} \geq \frac{\sqrt{2}}{4}\sqrt{T_3}d) = P(\vec{n} \geq \frac{\sqrt{2}}{4}\sqrt{T_3}d - \vec{s}_1)$$

$$S \in R: P(e) = P(\vec{s}_1)P(e | \vec{s}_1) + P(\vec{s}_2)P(e | \vec{s}_2)$$

$$= \frac{1}{2} (P(e | \vec{s}_1) + P(e | \vec{s}_2)) = P(e | \vec{s}_1)$$

$$= Q\left(\sqrt{\frac{d^2 s_1 s_2}{2N_0}}\right) = Q\left(\sqrt{\frac{5T_3 d^2}{4N_0}}\right)$$

$$= Q(\sqrt{rb})$$

Q2. Consider the baseband modulation with two equiprobable signals

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$$s_1(t) = g(t) \text{ and } s_2(t) = -g(t),$$

where $g(t) = \begin{cases} A, & \text{if } 0 \leq t < T_s; \\ 0, & \text{otherwise,} \end{cases}$ and T_s is the symbol duration. The signal is sent over a noisy

channel. At the receiver, after the matched filtering and sampling, the output signal is given by

$$r = s + w$$

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intensity = $\frac{1}{2}A$

where w is uniformly distributed within $[-a, a]$, with a being a positive constant. Assume w is independent of all the $s \in \{s_1, s_2\}$.

(a) What are the vector forms of $s_1(t)$ and $s_2(t)$? Find the average symbol energy E_s and the average bit energy E_b . Express them in terms of A and T_s . (6 marks)

(b) If $0 < a < A\sqrt{T_s}$, find a decision boundary for the receiver to minimize the bit error probability, and calculate the average bit error probability with the optimal detection rule. (14 marks)

GSD

$$P_b = P_e = P(e|s_i)$$

$$a) E_1 = E_2 = A^2 T_s$$

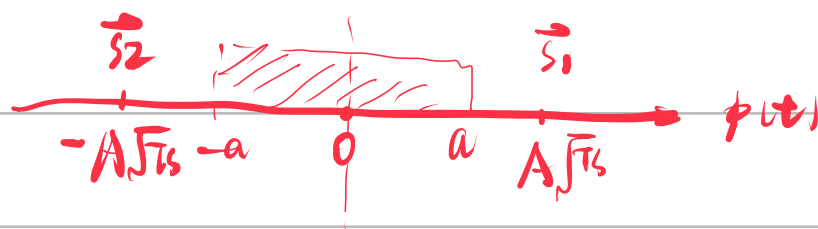
$$\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \frac{A}{A\sqrt{T_s}} = \sqrt{\frac{1}{T_s}}, \quad \vec{s}_1 = A\sqrt{T_s}$$

$$\vec{s}_2 = \langle s_2(t), \phi_1(t) \rangle = -A\sqrt{T_s}, \quad g_2(t) = s_2(t) - \vec{s}_2 \phi_1(t) = 0$$

$$E_s = \sum_{i=1}^2 p_i E_i = A^2 T_s$$

$$K = 1 \text{ bit} \rightarrow E_b = E_s = A^2 T_s$$

b)



$$ML: \hat{s} = \arg \max P(\vec{r} | \vec{s}_m)$$

$$\text{decision boundary: } \vec{r} = 0 \quad \begin{cases} > 0 & \vec{s}_1 \\ < 0 & \vec{s}_2 \end{cases}$$

$= -A\sqrt{T_0}$

$$P(t) = P(t | \vec{s}_1) = P(\vec{r} = \vec{s}_1 + \vec{w} < 0) = P(\vec{w} < -\vec{s}_1)$$
$$= 0?$$

